

STAT 153 & 248 - Time Series

Lecture Twelve

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In this lecture, we discuss the Bayesian analogue of ridge regularization. Let us start by recapping ridge regression from the previous lecture.

1 Recap: Ridge Regression

Our model from the last two lectures is given by:

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2 \text{ReLU}(t-2) + \cdots + \beta_{n-1} \text{ReLU}(t-(n-1)) + \epsilon_t \quad (1)$$

where, as always, $\epsilon_t \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$. Here $\text{ReLU}(t-c) = (t-c)_+$ equals 0 if $t \leq c$ and equals $(t-c)$ if $t > c$.

The unknown parameters in this model are $\beta_0, \beta_1, \dots, \beta_{n-1}$ as well as σ .

Alternatively, (1) can be written as:

$$y = X\beta + \epsilon$$

where

$$X = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 2 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & n-1 & n-2 & \cdot & \cdot & \cdot & 1 \end{pmatrix} \text{ and } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix}. \quad (2)$$

The ridge regression estimator $\hat{\beta}_{\text{ridge}}(\lambda)$ for β is given by the minimizer of:

$$\sum_{t=1}^n (y_t - \beta_0 - \beta_1(t-1) - \beta_2 \text{ReLU}(t-2) - \cdots - \beta_{n-1} \text{ReLU}(t-(n-1)))^2 + \lambda (\beta_2^2 + \beta_3^2 + \cdots + \beta_{n-1}^2). \quad (3)$$

The objective function above can also be written as

$$\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2.$$

It turns out that $\hat{\beta}_{\text{ridge}}(\lambda)$ can be written in closed form using matrix notation. To see this, note first that the gradient of the above objective function with respect to β is given by

$$\nabla \left(\|y - X\beta\|^2 + \lambda \sum_{t=2}^{n-1} \beta_t^2 \right) = -2X^T y + 2X^T X \beta + 2\lambda \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix}$$

Let J denote the $n \times n$ diagonal matrix whose diagonal entries are $0, 0, 1, \dots, 1$. In other words, the first two diagonal entries of J are 0 and the rest of the diagonal entries equal 1:

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix}$$

With this matrix, we can write

$$\nabla \left(\|y - X\beta\|^2 + \lambda \sum_{t=2}^{n-1} \beta_t^2 \right) = -2X^T y + 2X^T X \beta + 2\lambda \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix} = -2X^T y + 2X^T X \beta + 2\lambda J \beta.$$

Setting this gradient equal to zero, we get

$$-2X^T y + 2X^T X \beta + 2\lambda J \beta = 0 \implies (X^T X + \lambda J) \beta = X^T y.$$

which gives

$$\hat{\beta}_{\text{ridge}}(\lambda) = (X^T X + \lambda J)^{-1} X^T y. \quad (4)$$

This looks very similar to the usual linear regression least squares formula $(X^T X)^{-1} X^T y$ with the only difference being the presence of the λJ term.

2 Bayesian Regularization

We now treat regularization in high-dimensional regression from the Bayesian point of view. Before discussing regularization, let us first recap the basics of Bayesian regression in the model:

$$y = X\beta + \epsilon \quad \text{with } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

The basic prior that we used previously is

$$\beta_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-C, C).$$

for a large positive constant C . For this prior, we showed (see e.g., problem 5 in Homework 1) that

$$\beta \mid \text{data}, \sigma \sim N((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1}) \quad (5)$$

when $C \rightarrow \infty$. This fact is not quite true if C is not very large.

A slightly different prior which allows exact formulae even for finite C is the Gaussian prior:

$$\beta_j \stackrel{\text{i.i.d.}}{\sim} N(0, C). \quad (6)$$

Under this prior, it turns out that

$$\beta \mid \text{data}, \sigma \sim N\left(\left(\frac{X^T X}{\sigma^2} + \frac{I}{C}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + \frac{I}{C}\right)^{-1}\right) \quad (7)$$

where I is the identity matrix. It is instructive to compare (5) and (7). Unlike (5) which is only true for large C , the fact (7) is true for every $C > 0$. It is also clear that when $C \rightarrow \infty$, then (7) is the same as (5). Observe that when C is large, there is not much difference qualitatively between $\text{unif}(-C, C)$ and $N(0, C)$ (they are both uninformative priors).

We shall prove a more general form of (7) later in this lecture.

Now let us specialize to the case of the high dimensional regression (2). If we use the prior (6) with $C \rightarrow \infty$, then the posterior mean becomes the unregularized least squares (or unregularized MLE) estimator $(X^T X)^{-1} X^T y$. The fitted values will then perfectly interpolate the data leading to overfitting. From the Bayesian perspective, this is happening because the prior (6) with very large C is not useful for this dataset. The prior needs to be changed for a more meaningful analysis. In the frequentist analysis, the main motivation for the ridge regularization (3) is the need to obtain smaller estimates for $\beta_2, \dots, \beta_{n-1}$ which will lead to a smoother fit to the data. This same effect can be obtained by the following modification of the prior (6):

$$\beta_0, \beta_1 \stackrel{\text{i.i.d.}}{\sim} N(0, C) \quad \text{and} \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d.}}{\sim} N(0, \tau^2) \quad (8)$$

for a small parameter τ (in the above, we also assume that $\beta_0, \dots, \beta_{n-1}$ are all independent). The prior (8) can be written as

$$\beta \sim N(0, Q) \quad (9)$$

where Q is the diagonal matrix with diagonal entries $C, C, \tau^2, \dots, \tau^2$. Under the prior (9), the posterior of β is given by

$$\beta \mid \text{data}, \sigma \sim N\left(\left(\frac{X^T X}{\sigma^2} + Q^{-1}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + Q^{-1}\right)^{-1}\right) \quad (10)$$

We will prove this result later in this lecture. The posterior mean therefore is given by

$$\left(\frac{X^T X}{\sigma^2} + Q^{-1}\right)^{-1} \frac{X^T y}{\sigma^2} = (X^T X + \sigma^2 Q^{-1})^{-1} X^T y. \quad (11)$$

This expression is closely related to the ridge estimator (4). Note that Q^{-1} is diagonal with diagonal entries $1/C, 1/C, 1/\tau^2, \dots, 1/\tau^2$. When C is very large, the first two diagonal entries of Q^{-1} are very close to zero so that

$$Q^{-1} \approx \frac{1}{\tau^2} J.$$

Thus the posterior mean (11) is therefore

$$\left(X^T X + \frac{\sigma^2}{\tau^2} J\right)^{-1} X^T y$$

which matches (4) if

$$\lambda = \frac{\sigma^2}{\tau^2} \text{ or, equivalently } \tau = \frac{\sigma}{\sqrt{\lambda}}. \quad (12)$$

Ridge regularization therefore can be understood as Bayesian regression with the prior (8). The precise equivalence is obtained if λ is related to τ^2 via (12).

3 Bayesian approach for handling unknown λ

As we just described, the ridge regression estimate with regularization parameter λ can be interpreted as Bayesian regression with the prior (8) provided $\tau^2 = \sigma^2/\lambda$. In other words, ridge regression corresponds to the Bayesian prior:

$$\beta_0, \beta_1 \stackrel{\text{i.i.d.}}{\sim} N(0, C) \quad \text{and} \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2/\lambda). \quad (13)$$

In the frequentist framework, λ is chosen via some form of Cross-Validation. In the Bayesian framework, we can deal with λ being unknown by simply taking a prior on λ or on $\log \lambda$. One simple way of choosing this prior is via the following. We find two values $\lambda_{\min}$ and $\lambda_{\max}$ so that the ridge regression fit with $\lambda = \lambda_{\min}$ is nearly the same as the original data (overfitting) and so that the ridge regression fit with $\lambda = \lambda_{\max}$ is nearly the same as the simple line fit (underfitting). We then take

$$\log \lambda \sim \text{uniform}(\log \lambda_{\min}, \log \lambda_{\max}).$$

This is equivalent to:

$$\lambda \sim f_\lambda \quad \text{where} \quad f_\lambda(\lambda) = \frac{I\{\lambda_{\min} \leq \lambda \leq \lambda_{\max}\}}{\lambda (\log \lambda_{\max} - \log \lambda_{\min})} \propto \frac{I\{\lambda_{\min} \leq \lambda \leq \lambda_{\max}\}}{\lambda}$$

Also σ is also unknown parameter (as in any linear regression) so we place the usual prior on σ :

$$\log \sigma \sim \text{unif}(-C, C)$$

for a large C .

We can rewrite the whole prior as:

$$\beta \mid \lambda, \sigma \sim N(0, Q) \quad \text{and} \quad \lambda \sim f_\lambda \quad \text{and} \quad \log \sigma \sim \text{unif}(-C, C)$$

where Q is a diagonal matrix with diagonal entries $C, C, \sigma^2/\lambda, \dots, \sigma^2/\lambda$.

As in the case of previous models, we combine the prior with the likelihood to compute the posterior of β, λ, σ . If we integrate out β and σ from the posterior, we will obtain the posterior of λ alone. This will tell us all we need to know about which λ 's are good for this dataset.

We will carry out the calculations in the next lecture, but the answers will be given by (we take $C \rightarrow \infty$):

$$f_{\lambda|\text{data}}(\lambda) \propto f_\lambda(\lambda) \frac{\lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2}}{(y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y)^{(n/2)-1}} \quad (14)$$

and

$$\frac{1}{\sigma^2} \mid \text{data}, \lambda \sim \text{Gamma} \left(\frac{n}{2} - 1, \frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2} \right)$$

and

$$\beta \mid \text{data}, \lambda, \sigma \sim N \left((X^T X + \lambda J)^{-1} X^T y, \sigma^2 (X^T X + \lambda J)^{-1} \right).$$

These formulae can be used in many ways. For a simple way, one can compute (14) on a grid of $\log \lambda$ -values, then use it to estimate λ (e.g., via the posterior mean or median). Then one can just take the λ -estimate and use ridge regression. Alternatively, one can generate values of $(\lambda^{(i)}, \sigma^{(i)}, \beta^{(i)})$, $i = 1, \dots, N$ from the full posterior of (λ, σ, β) .